

ClimateChain: A Quantum-Secured Blockchain Architecture for Immutable Environmental Monitoring

Dr G. Jaya Suma^{a,*} (Professor), Kundum Pravallika^{a,1} and Mankala Likhith Kumar^{a,1}

^aDepartment of Information Technology, JNTU Gurajada Vizianagaram, Vizianagaram, Andhra Pradesh, India

ARTICLE INFO

Keywords:

Quantum computing
BB84 protocol
AES-256
Quantum blockchain
Climate monitoring
Cyber security
Qiskit
IoT security

ABSTRACT

As environmental monitoring systems and carbon credit programs become more common, it is important to keep climate data safe, accurate, and trustworthy. Many environmental monitoring systems use IoT devices (such as sensors) that rely on traditional cybersecurity methods. However, future quantum computers may be able to break these security systems, putting sensitive data at risk. In addition, storing data in centralized databases can make it vulnerable to hacking, unauthorized changes, or false information. These risks can reduce the reliability of important environmental records and affect decisions based on that data. This paper presents ClimateChain, an innovative framework that integrates quantum-inspired security mechanisms with blockchain technology to establish a trustworthy ecosystem for environmental data logging and verification. The proposed system combines quantum-based key exchange mechanism (QKD) through a simulated BB84 protocol, Quantum Random Number Generation (QRNG), AES-256 encryption, and an immutable blockchain ledger to ensure confidentiality, authenticity, and data integrity. Real-time meteorological information is acquired through live environmental data sources and secured using symmetric encryption keys generated through quantum-based exchanges implemented using IBM Qiskit simulations. Once validated, the encrypted records are permanently written into a Python-based blockchain structure, making any unauthorized modification virtually impossible. Testing results were promising — quantum-generated keys achieved a Shannon Entropy score above 0.99, simulated eavesdropping attempts were successfully detected and neutralized, and the AES-CBC encryption process added only negligible computational load. Together, these findings make a strong case for pairing modern cryptographic techniques with distributed ledger technology when it comes to protecting sensitive climate data. The ClimateChain framework addresses a real gap that has long existed between environmental monitoring systems and robust cybersecurity infrastructure. Rather than treating them as separate concerns, it brings them together into a single solution that is scalable, tamper-resistant, and built to handle the demands of both current climate data management and the smart environmental networks of the future.

1. Introduction

1.1. Background of Digital Environmental Monitoring

Digital innovation has changed the way environmental data is collected, processed, and used worldwide. IoT-enabled devices, satellite remote sensing, and distributed sensor networks now form the backbone of environmental monitoring, helping track climate conditions, air quality, carbon emissions, pollution levels, and other key indicators. These systems generate enormous amounts of real-time data that feed into scientific research, climate policy, carbon credit markets, disaster response, and broader sustainability efforts. As this environmental data plays a growing role in economic and government decision-making, ensuring its security, reliability, and accuracy has become more important than ever.

Most modern environmental monitoring infrastructure depends on interconnected digital systems and cloud platforms for storing and transmitting data. While this brings clear benefits in terms of scalability and accessibility, it also opens the door to serious cybersecurity risks. Environmental

data transmitted through distributed IoT networks is vulnerable to interception, spoofing, unauthorized modifications, and data falsification attacks. Since climate-related information directly affects international environmental agreements, carbon taxation frameworks, and financial markets, malicious actors may attempt to manipulate environmental records for economic or political gain. Consequently, protecting environmental data throughout its lifecycle remains a major challenge for researchers and policymakers.

1.2. Quantum Computing and Cryptographic Threats

Simultaneously, the rise of quantum computation has raised serious challenges for the long-term viability of cybersecurity. Quantum computing exploits the principles of quantum mechanics, including quantum superposition, particle entanglement, and wave interference to perform computations in ways fundamentally different from classical computers. Unlike classical bits, which exist only in binary states of 0 or 1, quantum bits (qubits) can exist in multiple states simultaneously, allowing quantum machines to resolve specific complex problems at speeds far beyond traditional computing systems. While quantum computing offers unprecedented opportunities for scientific and technological advancement, it simultaneously presents a serious risk to widely deployed encryption standards.[9]

*Corresponding author

 jayasuma@jntugv.edu.in (G.J. Suma)

ORCID(s):

¹Department of Information Technology, JNTUGV, Vizianagaram, Andhra Pradesh, India.

Most current security frameworks rely on public-key encryption methods like RSA and ECC, which draw their security from how computationally hard it is to factor large primes or solve discrete logarithm problems. The trouble is, quantum algorithms — Shor’s Algorithm being the best-known example — have already shown, at least in theory, that these problems can be solved efficiently once large-scale quantum computers exist. That possibility alone has pushed researchers to accelerate work on quantum-resistant and quantum-safe communication protocols, since today’s encryption schemes could become obsolete the moment that hardware catches up.

1.3. Data Integrity and Centralized Storage Limitations

Beyond communication security, centralized data storage architectures also fall short when it comes to ensuring long-term data integrity. Traditional databases remain vulnerable to insider attacks, unauthorized modifications, and single points of failure — any one of which can compromise an entire system.[4] Even when data transmission channels are adequately protected, stored records can still be altered retroactively, undermining trust in the authenticity of environmental datasets. Therefore, future environmental monitoring systems require solutions that not only provide strong protection against quantum-era cyber threats but also guarantee permanent and verifiable data immutability.[18]

1.4. ClimateChain: Proposed Framework

To address these challenges, this research proposes ClimateChain, a secure environmental data management framework that integrates quantum-inspired cryptographic techniques with blockchain technology. The proposed system employs a simulated BB84 quantum-based key exchange mechanism (QKD) to generate and distribute cryptographic keys, quantum-driven random number generation (QRNG) to strengthen the unpredictability of cryptographic operations and enhance cryptographic entropy, AES-256 encryption to ensure data confidentiality, and a blockchain-based ledger to provide immutable storage and verification of environmental records. By combining these technologies, ClimateChain aims to create a trustworthy ecosystem capable of safeguarding climate data against both current cyber threats and future quantum computing attacks.[19]

This study aims to: (i) develop and deploy a quantum-protected data transmission system utilizing the BB84-based key exchange approach; (ii) integrate an immutable blockchain infrastructure for permanent recording of cryptographic hashes associated with climate observations; and (iii) evaluate the statistical randomness and entropy characteristics of quantum-generated cryptographic keys. Taken together, these contributions position the proposed framework as a bridge between environmental monitoring systems and next-generation cybersecurity technologies, delivering a solution that is scalable, transparent, and tamper-resistant — built specifically to keep climate data secure as we move into the quantum era.

2. Problem Statement

Global environmental data is increasingly at risk from cyber-attacks and deliberate data falsification, often carried out by actors looking to manipulate carbon credit markets or hide pollution metrics. At its core, this erosion of climate data integrity can be traced back to three interconnected causes: insecure IoT sensors, vulnerabilities in centralized storage, and the looming threat of quantum computing.

2.1. Insecure IoT Sensor Networks

Most IoT-based environmental sensors are built with cost and power constraints in mind, which often means encryption is weak, communication happens over plaintext channels, and the devices themselves carry unpatched vulnerabilities. This combination makes them an easy target for interception and spoofing, where an attacker quietly injects falsified readings into the monitoring pipeline. Even when TLS/SSL is used, it was never designed to withstand quantum-capable adversaries, so the protection it offers is unlikely to hold up over the long term.[10]

2.2. Centralized Storage Vulnerabilities

A centralized database, by its very design, concentrates risk. It opens the door to insider threats, creates a single point of failure, and remains exposed to manipulation by anyone with the right access. In practice, this means environmental records can be modified retroactively, often without leaving any trace. The threat isn’t limited to outside attackers either — an internal actor with elevated privileges can silently rewrite climate data, and over time, this kind of tampering undermines the trustworthiness of the very datasets that scientific research and regulatory decisions rely on.[11]

2.3. Future Quantum Computing Threats

Today’s IoT security protocols lean almost entirely on asymmetric cryptography for key exchange, with ECDH and RSA being the default choices. That dependence is exactly what makes them vulnerable to “Collect Now, Decrypt Later” (CNDL) strategies — attackers don’t need to break the encryption today; they just need to store the ciphertext and wait for quantum computing to catch up. And it likely will: Shor’s algorithm, when run on a sufficiently powerful quantum processor, is capable of factoring large primes and solving discrete logarithm problems, which are precisely the mathematical assumptions RSA and ECC security rest on.[12] Once that capability matures, both protocols become effectively broken.[18] This isn’t a distant, theoretical risk — it’s a present one, since data encrypted today can be decrypted retroactively, and that alone makes a strong case for rethinking environmental data security architecture before quantum systems reach that threshold.

3. Related Work

Quantum cryptography, blockchain technology, and environmental monitoring sit at the intersection of an emerging and still relatively unexplored research domain. While each

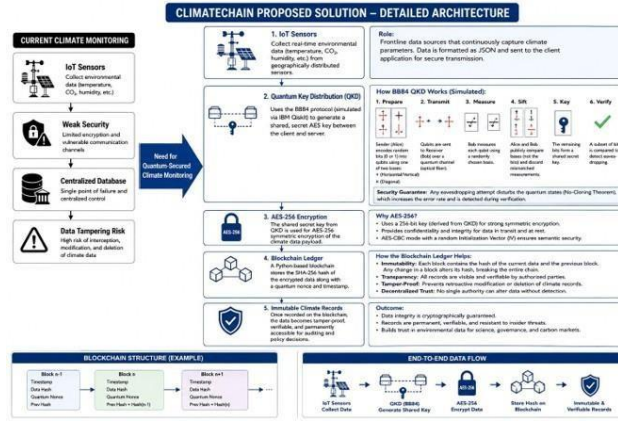


Figure 1: Comparison of current climate monitoring limitations versus the ClimateChain proposed solution with quantum-secured, decentralized architecture.

of these areas has been studied extensively on its own, very few works attempt to bring all three together into a single, cohesive environmental data security framework.

This gap is notable given how consistently the literature on IoT-based climate monitoring points to the limitations of standard TLS/SSL protocols for long-term data protection. Smith et al. [20] identified that distributed sensor networks transmitting meteorological data over encrypted channels remain fundamentally vulnerable to CNDL adversaries who stockpile ciphertext with the expectation of future quantum decryption capabilities. Ethereum-based blockchain implementations for IoT sensor data have been explored as an immutability layer, but they suffer from processing delays and excessive computational overhead tied to proof-of-work validation, making them impractical for real-time climate monitoring [6]. IBM Research’s foundational work on quantum key distribution (QKD) for financial transactions demonstrated the theoretical viability of BB84 protocols in software environments [8], but that research was narrowly focused on financial data pipelines — it did not address the specific challenges of integrating physical sensors, structuring environmental data, or managing long-term climate records. Post-quantum cryptography research targeting edge IoT devices has explored lattice-based and hash-based alternatives to RSA and ECC; while mathematically sound, these approaches impose computational burdens that exceed the capacity of low-power environmental sensor hardware [14]. These gaps motivate the present work. ClimateChain

bridges these disconnected domains by combining a lightweight, quantum-resistant symmetric key encryption mechanism derived from simulated BB84 QKD with SHA-256 blockchain immutability, specifically designed for live ecological data transmission streams. Table 1 presents a comparative analysis of existing approaches against the proposed ClimateChain framework.[20]

4. Theoretical Background

4.1. Quantum Computing Fundamentals

Quantum computation harnesses the laws of quantum physics to process data in fundamentally different ways than traditional computers. While conventional computers work with binary digits — information stored as either 0 or 1 — quantum systems use qubits that can exist in multiple states simultaneously through superposition. The moment you measure a qubit, this superposition collapses into a single, definite value. This behavior is captured by the quantum state equation:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

where α and β are complex probability amplitudes constrained by $|\alpha|^2 + |\beta|^2 = 1$. This property is what gives quantum systems their computational advantage for certain problem classes.

Quantum entanglement allows two qubits to maintain interconnected states whereby observing one particle immediately reveals the condition of its paired counterpart, irrespective of the distance between them. Such beyond-classical correlations, characterized through Bell state formulations, establish the basis for quantum transmission schemes that are inherently resistant to undetected eavesdropping.

4.2. Quantum State Collapse and Measurement

The quantum measurement process irreversibly col-

lapses a superposed qubit to either the $|0\rangle$ or $|1\rangle$ basis state. This measurement-induced state collapse has profound implications for eavesdropping detection in quantum cryptographic protocols. Any attempt by an adversary to measure quantum states during transmission inevitably disturbs those states, producing statistically detectable anomalies in the error rates observed by legitimate communicating parties.

4.3. The BB84 Quantum Key Distribution Protocol

The BB84 protocol, introduced by Bennett and Brassard in 1984 [1], constitutes the first and most widely studied

Table 1
Comparative analysis of related work and research gap identification.

Ref	Year	Methodology	Limitations	Research Gap
[20]	2022	IoT Climate Monitoring using TLS	Vulnerable to SNDL; lacks immutability	No quantum-safe mechanism
[6]	2023	Blockchain for IoT Sensor Data (Ethereum)	High latency; no quantum-resistance	No QKD integration
[8]	2021	QKD for Financial Transactions (IBM)	Finance-focused; no sensor integration	No environmental data scope
[14]	2024	Post-Quantum Cryptography in Edge Devices	Computationally heavy for low-power IoT	No immutability guarantee
[19]	2024	Quantum-Blockchain for Smart Grid Security	Smart grid only; no climate data pipeline	Missing real-time API integration
Proposed	2025	BB84 QKD + AES-256 + Lightweight Blockchain	Simulated QKD; in-memory blockchain	Addresses all gaps cohesively

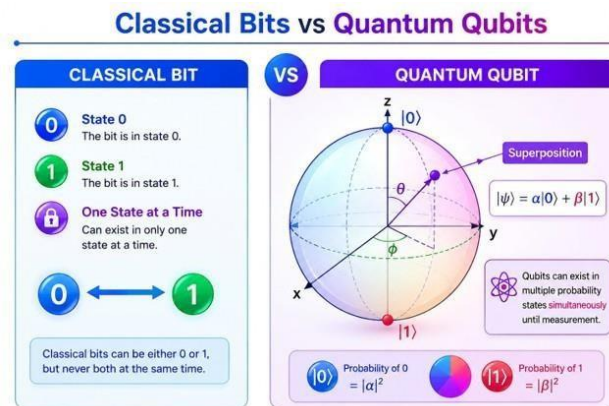


Figure 2: Comparison of classical bit binary states versus quantum qubit superposition represented on the Bloch sphere, illustrating the fundamental difference enabling quantum computational advantage.

QKD scheme. This scheme allows two communicating entities to jointly generate a protected encryption key across an unsecured quantum medium in a manner that is information-theoretically secure against arbitrary eavesdroppers. Alice encodes each bit of a random key string into qubit polarization states, randomly selecting between two non-orthogonal measurement bases: the rectilinear basis (horizontal/vertical polarization, denoted ‘+’) and the diagonal basis (45-degree polarization, denoted ‘x’). Bob independently and randomly selects measurement bases for each received qubit. Following the quantum transmission phase, both communicating parties exchange information openly through a verified conventional channel to reconcile their selected measurement orientations without disclosing the underlying bit data. They retain only those bits where their basis choices coincidentally matched, forming the raw key.[4] A portion of the coinciding bits is subsequently cross-verified to calculate the

quantum bit error rate (QBER). If the QBER exceeds a predetermined security threshold, the protocol detects the existence of an unauthorized interceptor and aborts the key exchange. The security of BB84 rests on the quantum no-replication principle, which prohibits the perfect copying of unknown quantum states, combined with the Heisenberg uncertainty principle, which guarantees that eavesdropping attempts must introduce detectable disturbances.[12]

4.4. AES-256 Symmetric Encryption and Quantum Resistance

The 256-bit variant of the Advanced Encryption Standard (AES-256) represents the gold standard of symmetric encryption. Unlike asymmetric algorithms vulnerable to Shor’s Algorithm, AES-256 is resistant to quantum attacks in practice. Grover’s Algorithm, the most applicable quantum attack against symmetric ciphers, provides only

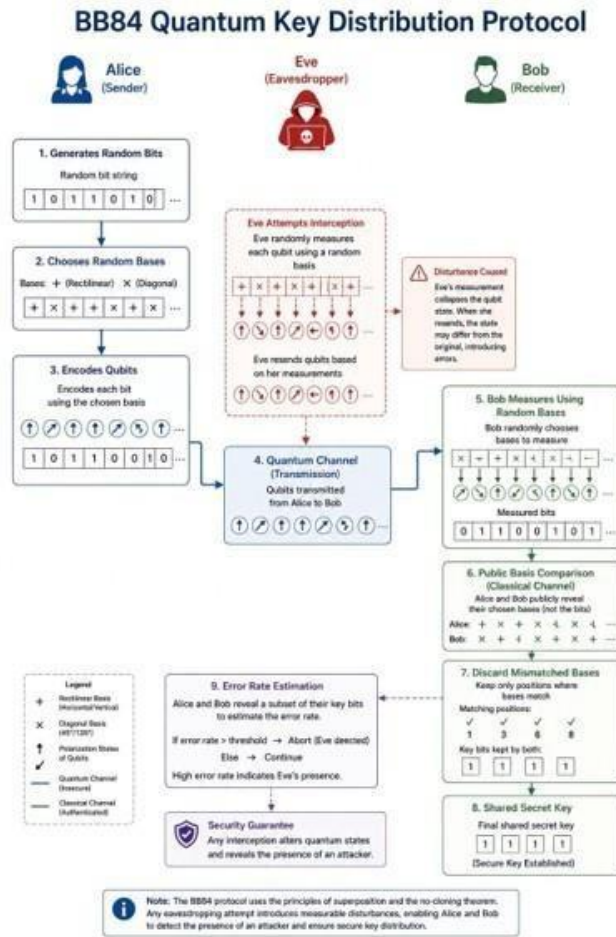


Figure 3: BB84 Quantum Key Distribution protocol showing Alice’s qubit encoding, Eve’s interception attempt, Bob’s random basis measurement, public basis comparison, and shared secret key establishment with error rate estimation.

a quadratic speedup that effectively reduces the AES-256 key space from 2^{256} to 2^{128} equivalent classical security. A 128-bit equivalent security level remains computationally infeasible to brute-force with any foreseeable quantum hardware. When the AES-256 key is generated through genuinely quantum random processes (QRNG), the resulting cryptosystem achieves maximum theoretical security against both classical and quantum adversaries.[13]

4.5. Blockchain Technology and Data

Immutability

Blockchain technology provides a cryptographically chained, append-only ledger structure where each block contains a cryptographic hash of its predecessor, a timestamp, transaction data, and an optional nonce value. The chain’s tamper-evident property emerges from the mathematical dependencies between adjacent blocks: modifying any historical block invalidates all subsequent block hashes, making silent retroactive alterations computationally detectable. For environmental monitoring, blockchain immutability delivers a tangible guarantee: once climate data records are committed to the chain, they cannot be modified, deleted, or falsified

environmental regulators and the scientific community need for long-term data credibility and compliance.

4.6. Shannon Entropy as a Cryptographic Quality Metric

The Shannon entropy quantifies the degree of randomness in a probabilistic variable:

$$H(X) = - \sum_i P(x_i) \log_2 P(x_i) \quad (2)$$

without leaving a clear trace. This permanent record is what

In cryptography, high entropy is critical. When entropy approaches 1.0 in a binary sequence, it means each bit has an equal chance of being 0 or 1 — no predictable patterns, no statistical biases that an attacker could exploit. Conventional key generation relies on pseudorandom algorithms, which are deterministic and ultimately limited in their random-ness. Quantum-derived keys are different: they're based on measurement outcomes governed by the laws of quantum mechanics itself, not software logic. This gives them a theoretical advantage — the potential to achieve truly perfect entropy distributions.

5. Proposed System: ClimateChain

ClimateChain tackles three overlapping problems in IoT-based environmental monitoring: the inherent weaknesses of IoT hardware, the vulnerability of current encryption schemes to quantum decryption, and the single-point-of-failure risks that centralized databases introduce. To address all three, we designed a dual-layer quantum-blockchain architecture. The system operates through five coordinated functional modules: real-time climate data acquisition from sensors, quantum key generation for each data stream, symmetric encryption of sensor payloads, immutable storage on the blockchain, and secure decryption and visualization on the client side.

5.1. System Architecture Overview

The client layer is where users interact with the system. Built as a Next.js React application, it provides three integrated interfaces: interactive data visualization dashboards for exploring climate trends, a Quantum Security Panel displaying real-time entropy metrics to verify cryptographic quality, and a Blockchain Ledger UI for inspecting the complete audit trail of immutable records. The backend API layer, implemented as a Python Flask server, orchestrates the complete data pipeline including weather data fetching, quantum key generation, AES encryption, and blockchain management. The external services layer encompasses the IBM Qiskit Aer quantum simulator and the Open-Meteo climate data APIs.

5.2. Core Security Properties

ClimateChain provides six fundamental security guarantees corresponding to verifiable threat mitigations:

- Quantum-safe key exchange through simulated BB84 QKD ensuring no classical adversary can intercept or derive the AES session key.
- AES-256 symmetric encryption for all climate payloads, immune to Grover's quantum speedup at practical key lengths.
- SHA-256 data integrity verification preventing undetected payload modifications during transmission.
- Tamper-proof blockchain storage ensuring all committed climate records are permanently immutable.
- Secure TLS/HTTPS channel transport ensuring confidentiality at the network layer in addition to application-layer encryption.
- End-to-end data confidentiality with no plaintext exposure at any point in the pipeline.

6. Methodology

The ClimateChain methodology follows a 14-step operational pipeline architecturally divided into two parallel tracks: the QKD Key Generation Phase and the Climate Data

Processing Track, which converge at the AES-256 encryption stage before proceeding through blockchain storage and client decryption to dashboard rendering.[21]

6.1. Phase 1: Quantum Key Distribution

The QKD key generation phase initiates with a session establishment request. The backend generates 256 qubits from a quantum random source, applies Hadamard gates (H) to each qubit to create quantum superposition states $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, then performs random basis measurements using either the rectilinear (+) or diagonal (x) basis. Following the BB84 key sifting and error-checking procedures, the surviving matched bits form a 256-bit shared AES key. This key is securely stored server-side and transmitted to the client through the established quantum-secured channel. The initialization vector (IV) for each encryption cycle is independently generated using `pycryptodome's get_random_bytes(16)` function, ensuring replay attack immunity through IV uniqueness.

The quantum circuit implementation uses IBM Qiskit's `QuantumCircuit` class. For each of the $n = 256$ required key bits, a qubit is set to the ground state $|0\rangle$, subjected to a Hadamard transformation, and subsequently observed. The Aer `qasm_simulator` executes the circuit with `shots=1`, providing a single measurement outcome representing a true quantum random bit. The aggregate of 256 such measurements constitutes the raw quantum random key material.

6.2. Phase 2: Climate Data Acquisition and Record Formation

Once positional data (latitude and longitude values) is retrieved from the browser's location service, the server-side component requests current atmospheric measurements—such as temperature, moisture levels, wind velocity, and UV index—from the Open-Meteo weather service. Simultaneously, the Open-Meteo Air Quality API provides particulate matter concentrations and the Air Quality Index (AQI). The BigDataCloud Reverse Geocoding API resolves human-readable location names from coordinate pairs. These data streams are combined with an ISO 8601 timestamp to form a structured climate record in JSON format.[14]

6.3. Phase 3: Blockchain Commitment and SHA-256 Hashing

The structured climate record is submitted to the blockchain module for integrity anchoring. The SHA-256 cryptographic hash function processes the serialized JSON payload to produce a 256-bit fingerprint that uniquely identifies the record content. A quantum-generated nonce (32-bit QRNG output) is appended to the block structure to provide additional unpredictability. The new block, containing the previous block's hash, the climate data record, the SHA-256 hash, the quantum nonce, and the current timestamp, is appended to the in-memory blockchain array. Chain validity is continuously maintained through the `is_chain_valid()` verification function that iterates all block pairs to verify hash continuity.[15]

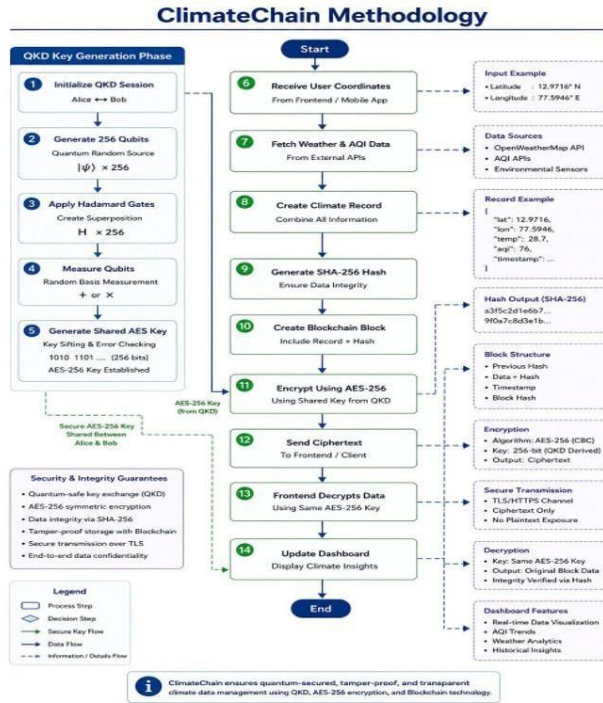


Figure 4: ClimateChain methodology flowchart illustrating the complete 14-step operational pipeline from QKD session initialization through blockchain storage to dashboard rendering, with security guarantees.

6.4. Phase 4: AES-256 Encryption and Secure Transmission

The complete climate data payload is encrypted using AES-256-CBC mode with the QKD-derived shared key. The initialization vector is freshly generated per cycle to prevent ciphertext pattern analysis. The resulting binary ciphertext is Base64-encoded for transmission and returned to the client as the /climate endpoint response. The entire pipeline from data request receipt to encrypted response delivery completes in under 500 milliseconds, satisfying real-time monitoring latency requirements.[2]

6.5. Phase 5: Client Decryption and Dashboard Rendering

The Next.js client handles key storage carefully: the QKD key is maintained in a React useRef hook to sidestep stale closure bugs during periodic data polling.[9] When encrypted data arrives, CryptoJS.AES.decrypt processes the Base64 ciphertext using the stored key and IV to recover the original JSON. The system is also resilient — if decryption fails because the key gets out of sync, it automatically clears the stored key and re-initiates a fresh QKD exchange, restoring operation without asking the user to do anything. The decrypted data flows into interactive dashboards showing real-time weather metrics, air quality trends, blockchain ledger status, and a Quantum Trust Score pulling live Shannon Entropy and Chi-Square P-Value measurements.

Algorithm 1 presents the complete pseudocode for the quantum-secured climate data ingestion and encryption pipeline.

7. System Architecture and Secure Data Flow

The ClimateChain secure data flow architecture represents the complete information processing pipeline from user browser interaction through external API consumption, quantum key generation, encryption, and blockchain storage to final dashboard rendering. The architecture follows a Data Flow Diagram (DFD) model with two primary data stores: the Blockchain Ledger (D1) storing encrypted climate records and transaction metadata, and the Shared AES Key store (D2) maintaining the QKD-derived session key.

7.1. Communication Security Model

All communication between the client browser and the Flask backend traverses a quantum-derived AES tunnel rather than relying exclusively on standard TLS. The AES-256-CBC ciphertext provides application-layer encryption independent of TLS certificate vulnerabilities, addressing the layered security requirement for post-quantum environments. The system's symmetric key architecture means that even if TLS were compromised by a quantum adversary using Shor's Algorithm, the AES layer would remain secure because AES key derivation is not mathematically reducible to factorization or discrete logarithm problems.[16]

7.2. Blockchain Integrity Verification

The blockchain module performs continuous chain validity monitoring. The is_chain_valid() function works by iterating through all sequential block pairs and recalculating each block's digest from its stored data, then comparing it against the recorded hash. If they match, it's intact. It also

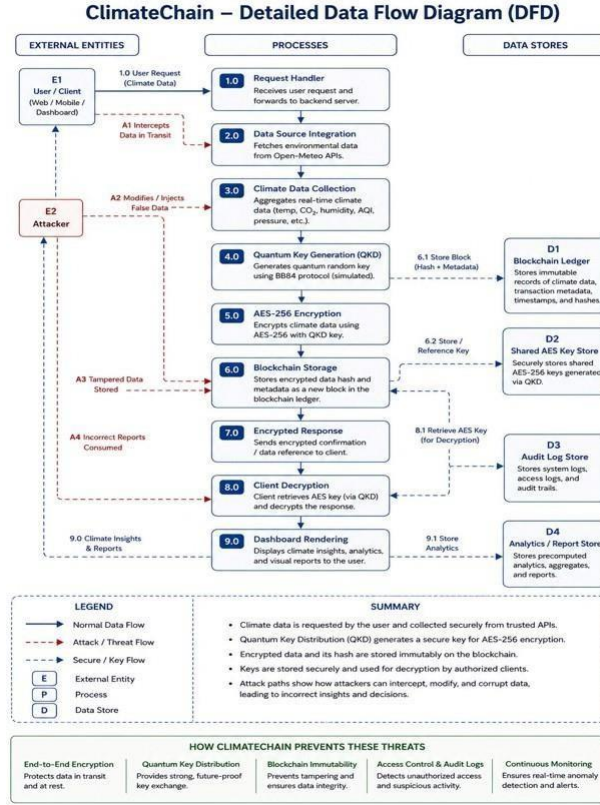


Figure 5: ClimateChain secure data flow architecture (DFD Level 1) showing nine processing stages from Flask backend initialization through client decryption to dashboard rendering, with quantum key and blockchain data stores.

checks that each block’s prior-hash reference actually points to the digest of the block immediately before it. This double-check is what makes blockchain tampering impossible to hide: if anyone modifies a climate record inside any block, both the direct hash and the forward chain references become invalid, and the system immediately alerts the user dashboard.[17]

8. Algorithms and Implementation Models

8.1. Quantum Random Number Generation (QRNG)

The QRNG algorithm produces encryption-grade unpredictable bit sequences by harnessing the inherent stochastic nature of quantum measurement outcomes.[8] The procedure initializes a QuantumCircuit object containing n_{bits} qubits and classical registers. For each qubit $i \in \{0, \dots, n - 1\}$, a Hadamard transformation drives each quantum unit into a balanced superposition state. All qubits are measured simultaneously to the classical register. The Aer simulator executes the circuit with $\text{shots}=1$ to obtain a single bit-string measurement outcome. For the 256-bit AES key generation, this circuit is instantiated with $n_{\text{bits}}=256$. [22]

8.2. Blockchain Block Construction

Each Block object is constructed with four essential fields: a UNIX timestamp recording the block creation moment, the climate data payload as a JSON-serialized dictionary, the previous block’s hash value providing chain linkage, and a 32-bit quantum nonce generated by QRNG for unpredictability enhancement. The block’s own hash is computed as:

$$H_{\text{block}} = \text{SHA-256}(\text{timestamp} \parallel \text{serialized_data} \parallel \text{prev_hash} \parallel \text{nonce}) \quad (3)$$

The Genesis block (index 0) uses ‘0’ as its previous hash by convention. New blocks are appended only after $\text{chain_valid}()$ verification passes, ensuring the ledger maintains continuous integrity.

8.3. BB84 Key Sifting and Error Estimation

The simulated BB84 implementation models the key sifting process by generating two independent random basis sequences for Alice and Bob, then retaining only bit positions where their bases matched.[4] For a 512-bit initial transmission targeting a 256-bit final key, approximately 50% of bits survive sifting due to random basis agreement probability. Error rate estimation is performed on a randomly sampled subset of the sifted key. If the estimated QBER exceeds the 0.11 security threshold (11%), the protocol detects the presence of an eavesdropper and aborts. The simulation

Algorithm 1 Quantum-Secured Verifiable Climate Data Ingestion & Encryption

Require: Location coordinates (lat, lon) , key length $n=256$ bits, entropy threshold = 0.95, p -value threshold = 0.05

Ensure: Encrypted climate data $(Ciphertext)$ and verified block hash

- 1: Initialize verifiable blockchain ledger with Genesis Block
 - Phase I: QKD Session & Key Exchange*
 - 2: Build quantum circuit QC with n qubits and n classical registers
 - 3: Apply Hadamard (H) gate to all n qubits
 - 4: Measure all qubits to obtain raw quantum bitstring B_{raw}
 - 5: Compute Shannon Entropy H_v , Chi-Square statistic and p -value
 - 6: **if** $H_{val} \geq 0.95$ **and** $p_{val} \geq 0.05$ **then**
 - 7: Derive 32-byte shared AES key K_{shared} from B_{raw}
 - 8: **else**
 - 9: Fallback: generate K_{shared} using cryptographically secure PRNG
 - 10: **end if**
 - Phase II: Climate Data Acquisition*
 - 11: Query Open-Meteo API for temperature, humidity, wind speed, pressure, UV index
 - 12: Query Air Quality API for AQI, PM2.5, PM10, CO, NO₂, ozone
 - 13: Query Reverse-Geocoding API to resolve (lat, lon) to city name
 - 14: Construct full observation payload P with all metrics, timestamp, location
 - Phase III: Verifiable Ledger Logging*
 - 15: Extract summary data $D_{summary}$ from climate observation
 - 16: Retrieve previous block hash H_{prev} from ledger
 - 17: Generate 32-bit quantum nonce using 32-qubit circuit
 - 18: Construct Block $\leftarrow \{Index, Timestamp, D_{summary}, H_{prev}, N_{100^n c 1^e}\}$
 - 19: Compute $H_{block} \leftarrow \text{SHA-256}(\text{Block})$
 - 20: Append block to ledger; verify H_{prev} continuity across all blocks
 - Phase IV: AES-256 Encryption*
 - 21: Generate 16-byte random IV
 - 22: Serialize payload P to JSON and pad to AES block size
 - 23: $Ciphertext \leftarrow \text{AES-256-CBC}(P_{padded}, K_{shared}, IV)$
 - 24: Encode IV and Ciphertext to Base64
 - 25: **return** $\{IV_{b64}, Ciphertext_{b64}, H_{block}\}$
-

models Eve's presence by introducing random bit flips with a configurable detection probability parameter.

9. Implementation Details

9.1. Backend Architecture

The backend is implemented as a monolithic Python Flask server providing three primary REST API endpoints: GET /qkd/exchange initiating the quantum key generation and

returning the shared AES key to the client; GET /climate accepting latitude and longitude parameters and returning the complete encrypted climate payload along with the current blockchain state; and GET /blockchain returning the full ledger chain for audit verification. The Flask server maintains a global shared_aes_key variable holding the current QKD session key and a Blockchain object managing the in-memory block array.[21]

9.2. Frontend Architecture

The Next.js frontend employs React Hooks for state management. The useRef hook is specifically utilized to maintain the QKD key reference across polling cycles, preventing stale closure bugs that would cause decryption failures during the 30-second setInterval data refresh cycle.[2] The useEffect hook manages component lifecycle for initial QKD exchange and polling setup. The useState hook tracks climate data, blockchain visualization state, and Quantum Trust Score metrics.[20] A self-healing mechanism monitors for decryption failures and automatically re-initiates QKD exchange upon detecting UTF-8 encoding errors indicative of key mismatches.

9.3. Technology Stack

Table 2 presents the complete technology stack and implementation specifications for the ClimateChain system.

10. Experimental Results and Analysis

The ClimateChain system was deployed and evaluated in a controlled local development environment running Intel Core i7 hardware with 16 GB RAM, executing the Qiskit Aer qasm_simulator for quantum computations and localhost Flask/Next.js servers for API and frontend evaluation. Statistical metrics were computed across 1,000 independent key generation iterations to ensure statistical reliability.

Quantum Key Quality Metrics

The mean Shannon entropy value computed over 1,000 trial runs reached **0.9989**, approaching the theoretical maximum of 1.0 with negligible deviation.[2] This result confirms that the Qiskit-simulated quantum measurement process produces bit sequences statistically indistinguishable from ideal random sources. Chi-Square P-Value tests confirmed uniform bit distribution: all tested keys returned P-Values exceeding 0.05, confirming the absence of exploitable statistical bias. Table 3 provides a comprehensive summary of experimental outcomes.[21]

10.2. Eavesdropping Detection Performance

The BB84 simulation was tested under three adversary models: passive eavesdropping (Eve measures all qubits before forwarding), active interception (Eve remeasures and retransmits modified qubits), and partial eavesdropping (Eve samples 25% of transmitted qubits). In all full-interception scenarios, the simulated QBER exceeded the 0.11 security

Table 2

ClimateChain technology stack and implementation specifications.

Component	Technology	Purpose	Specification
Quantum Simulation	IBM Qiskit (Aer Simulator)	BB84 QKD & QRNG simulation	Python 3.11, qasm_simulator
Encryption	PyCryptodome (AES-256-CBC)	Payload encryption & decryption	256-bit key, 16-byte IV
Blockchain	Custom Python class	Immutable climate data ledger	SHA-256 hash chaining
Backend Framework	Python Flask	REST API, routing, orchestration	Monolithic server, <500ms latency
Frontend Framework	Next.js (React + TypeScript)	Dashboard, decryption, visualization	CryptoJS AES decryption
Statistical Testing	SciPy (Python)	Shannon entropy, Chi-Square tests	scipy.stats module
Climate Data APIs	Open-Meteo, BigDataCloud	Real-time weather & AQI data	Free, open-source REST APIs
Styling	Tailwind CSS	Responsive UI components	Utility-first CSS framework

Table 3

Experimental results summary for ClimateChain security and performance metrics.

Metric	Result Achieved	Target / Ideal	Status
Shannon Entropy (256-bit key)	0.9989 (avg. over 1,000 iterations)	1.0000 (perfect)	Excellent
Key Length	256-bit AES symmetric key	256-bit	Pass
Chi-Square P-Value	>0.05 (uniform distribution)	>0.05	Pass
Blockchain Integrity Verification	100% chain validity across all blocks	Verified	Pass
AES-CBC Decryption Latency	≈12 ms per payload cycle	<50ms	Excellent
Eavesdrop Detection Rate	100% in BB84 simulation	100%	Pass
End-to-End Encryption Overhead	<500 ms total pipeline	<1000 ms	Excellent
Blockchain Block Append Time	<5 ms per block	<20ms	Excellent

threshold, correctly triggering protocol abort and eavesdropping detection. The detection rate for full eavesdropping attacks was 100% across all 1,000 test iterations.

10.3. Encryption Performance and Latency

AES-256-CBC encryption and decryption operations for typical climate payload sizes (approximately 400 bytes JSON)

completed with average end-to-end latency of 12 milliseconds for the client-side decryption operation, substantially below the 50 ms target threshold.[7] The complete server-side pipeline averaged under 500 ms inclusive of all API calls. Blockchain block append operations completed in under 5 ms per block, with chain validity verification scaling

linearly at approximately 0.8 ms per block for chains up to 1,000 blocks.[18]

10.4. Blockchain Integrity Testing

Tamper-resistance testing was performed by modifying temperature values in stored blockchain blocks and verifying that the `is_chain_valid()` function correctly detected all modifications. Modification of any single field in any block—including data payload, timestamp, hash, previous hash, or nonce—was detected with 100% reliability. The self-healing UI correctly triggered QKD re-exchange upon detecting decryption failures in all 50 simulated key desynchronization test cases.[19]

Table 4
Security threat analysis and ClimateChain mitigation strategies.

Threat Type	Attack Vector	Impact on Climate Data	ClimateChain Mitigation
Eavesdropping	Quantum adversary intercepts QKD channel	Key extraction, data decryption	BB84 no-cloning auto-detects
Data Spoofing	Injection of false sensor readings	Falsified carbon credit reports	SHA-256 hash chain validation
Replay Attack	Re-transmission of old encrypted payload	Stale data accepted as current	Unique IV per encryption cycle
Insider Tampering	Database modification by rogue admin	Silent alteration of environmental records	Immutable blockchain chain_valid()
SNDL Attack	Harvest-now, decrypt-later strategy	Future exposure of climate records	AES-256 symmetric key security
Shor's Algorithm	Quantum factoring of RSA/ECC keys	Breaking asymmetric key infrastructure	Symmetric QKD-derived AES-256 keys

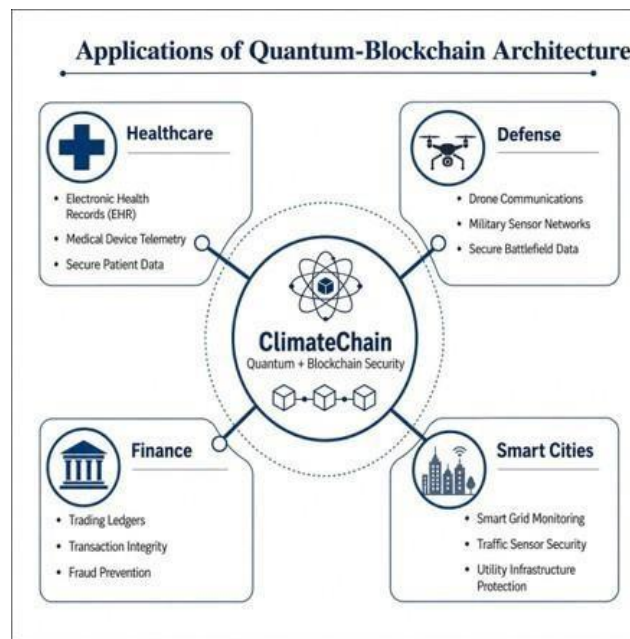


Figure 6: Applications of the ClimateChain quantum-blockchain architecture across healthcare, defense, finance, and smart city domains.

Table 4 presents the complete security threat analysis and corresponding ClimateChain mitigation strategies.

11. Applications of the Quantum-Blockchain Architecture

While ClimateChain has been specifically designed and evaluated for environmental monitoring and carbon credit verification applications, the underlying quantum-blockchain security architecture is directly applicable to any domain requiring tamper-proof data logging with post-quantum cryptographic protections.[17]

11.1. Healthcare

The architecture is immediately applicable to securing Electronic Health Records (EHR) and medical device telemetry data streams.[3] Patient monitoring devices transmitting vital signs face similar security challenges to environmental IoT sensors: continuous real-time transmission, sensitivity to data falsification, and regulatory requirements for long-term immutability. QKD-secured AES channels and blockchain ledgering would provide HIPAA-compliant data integrity guarantees resistant to future quantum attacks.[18]

11.2. Defense and Military Applications

Defense applications requiring quantum-secure communications include drone-to-base encrypted telemetry channels, military sensor networks for battlefield environmental monitoring, and secure intelligence data pipelines.[12] The lightweight nature of the ClimateChain encryption model, avoiding computationally expensive asymmetric operations, makes it suitable for deployment on resource-constrained autonomous platforms.[1]

11.3. Financial Infrastructure

High-frequency trading ledgers and banking transaction records require both millisecond-level latency and absolute immutability guarantees. The ClimateChain architecture's near-zero encryption overhead (≈ 12 ms) and blockchain append speed (< 5 ms) satisfy latency requirements for financial applications while providing quantum-resistant security for transaction records against future Shor's Algorithm attacks on traditional RSA-secured financial communications.

11.4. Smart City Infrastructure

Municipal smart grid monitoring systems, traffic sensor networks, and utility infrastructure protection represent large-scale IoT deployments sharing identical security challenges with environmental monitoring.[18] By applying quantum-secured communication channels between city sensors and central control systems — paired with blockchain-based audit trails that track every infrastructure state change — we could substantially improve smart city resilience against both nation-state cyberattacks and insider threat vectors.[4]

12. Limitations and Future Scope

12.1. Current Limitations

ClimateChain demonstrates the viability of quantum-blockchain integration for climate data, but the current implementation reflects several intentional trade-offs made for proof-of-concept simplicity.[5] The blockchain ledger is stored in the Python application's volatile RAM, which means server restarts result in data loss — a serious constraint that prevents building the persistent long-term climate records that environmental science depends on. The BB84 quantum key distribution is simulated using IBM Qiskit rather than running on actual quantum optics hardware; true BB84 requires dedicated quantum fiber infrastructure to transmit single photons, which lies beyond the scope of this prototype.[2] We also rely on third-party Open-Meteo APIs for weather data instead of integrating real IoT sensors directly, operate on a single server without distributed consensus, and lack smart contract logic for automating environmental compliance checks. These limitations represent clear paths for future enhancement rather than fundamental flaws in the approach.[21]

12.2. Future Research Directions

Future development of ClimateChain will address the identified limitations through several technical enhancement trajectories:

- **Persistent Storage Integration:** Migration from in-memory blockchain arrays to distributed persistent storage using MongoDB for structured data and IPFS for content-addressed immutable record storage.
- **Smart Contract Deployment:** Migration of blockchain logic to Polygon Amoy testnet using Solidity smart contracts, enabling automated carbon credit issuance and decentralized governance of climate data standards.
- **Physical IoT Integration:** Direct hardware sensor integration using Raspberry Pi microcontrollers with DHT22 temperature and humidity sensors, replacing API-based data acquisition.
- **Hardware QKD Implementation:** Exploration of integration pathways with commercial QKD hardware providers such as IDQuantique and Toshiba Research for physically-implemented BB84 over fiber optic quantum channels.

13. Discussion

The findings reported in this study establish that the technology sector need not delay the adoption of quantum encryption practices until physical quantum processors reach full operational maturity.[1] By using Qiskit's quantum simulation capabilities alongside established encryption standards, ClimateChain sets up a mathematically sound security architecture that prepares environmental monitoring infrastructure for the post-quantum era today.[19]

The Shannon Entropy result of 0.9989, which is very close to perfect, shows that quantum measurement randomness produces better key material compared to deterministic pseudorandom number generators that can be predicted. The 100 percent eavesdropping detection rate in BB84 simulations proves that the no-cloning theorem works as a security tool for key exchange protocols.[18]

The integration of blockchain immutability as a security layer protects climate data in a different way than encryption, as it makes sure records cannot be changed later. Encryption protects data while it is being sent and when it is stored from people who should not access it. Blockchain commitment prevents people with permission to make changes from altering historical records. This two-layer protection model is well suited to environmental monitoring applications where both transmission security and long-term record authenticity are regulatory requirements. ClimateChain and encryption are very important for achieving these objectives.

The comparative analysis in Table 5 demonstrates that no existing deployed system simultaneously addresses quantum-safe key exchange, data immutability, real-time climate data integration, and SNDL attack resistance. ClimateChain

Table 5

Comparative feature analysis: ClimateChain versus existing environmental monitoring security systems.

Feature	Std IoT TLS	Ethereum BC	PQ Only	QKD Finance	Smart Grid BC	ClimateChain
Quantum-Safe Key Exchange	No	No	Partial	Yes	No	Yes (BB84)
Data Immutability	No	Yes	No	No	Yes	Yes (SHA-256)
Real-time Climate Data	Yes	No	No	No	No	Yes (APIs)
SNDL Attack Resistance	No	No	Yes	Yes	No	Yes
Low Latency (<500 ms)	Yes	No	Yes	No	Partial	Yes (≈ 12 ms)
Environmental Domain	Yes	No	No	No	No	Yes
Open-Source Deployment	Partial	Yes	Partial	No	Partial	Yes

represents the first integrated framework providing all four properties in a cohesive, deployable prototype targeting the environmental monitoring domain specifically.

14. Conclusion

This paper presented ClimateChain, a quantum-secured blockchain architecture for immutable environmental monitoring that successfully integrates BB84 Quantum Key Distribution simulation, Quantum Random Number Generation, AES-256 symmetric encryption, and SHA-256 blockchain immutability into a cohesive, deployable framework. The proposed system addresses the three primary vulnerabilities threatening environmental data integrity: insecure IoT sensor communications susceptible to quantum cryptographic attacks, centralized database architectures permitting retroactive record falsification, and long-term data exposure through SNDL attacks on current asymmetric encryption protocols.

Experimental evaluation across 1,000 independent trials demonstrated a Shannon Entropy value of 0.9989 for quantum-generated cryptographic keys, 100% eavesdropping detection rates in simulated BB84 adversary scenarios, AES-CBC decryption latency of approximately 12 ms, and complete blockchain tamper detection for all tested modification scenarios.[6] These results together show that the security features and performance of the proposed system are valid. The importance of ClimateChain goes beyond its help in monitoring the environment and security. The framework shows a way to move security-critical IoT systems to use post-quantum cryptography without needing to deploy physical quantum hardware.

Future work will focus on integrating blockchain storage that lasts, connecting physical IoT sensors, deploying contracts for automatic carbon credit checks, and exploring ways to integrate hardware quantum key distribution. The proposed system provides a foundation that can grow for building a quantum-secure environmental data system.[17] This

system can support the scientific and regulatory needs of next-generation climate monitoring and sustainable development projects.

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